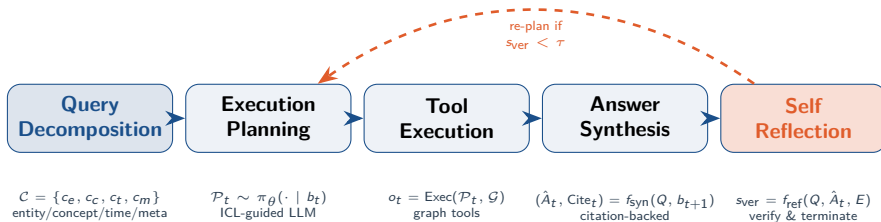


GRGA Planning: A Cognitive “Search-Reason-Verify” Loop



Action Space (4 Graph Tools):

- **Semantic Search:** retrieve node text
- **Temporal Filter:** constrain time ranges
- **Graph Traversal:** follow speaker/semantic links
- **Audio Access:** inspect acoustic cues

Key Design Choices:

- **Training-free:** LLM policy via in-context learning
- **Adaptive depth:** shallow for factual, deeper for temporal/inferential
- **Self-correction:** reflection filters low-evidence answers